	AzureSky	
	ISO Volatile Matter Method Validation	Document no: ASMV 001
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**
	Approved by:	**Nonhlanhla Msibi - Operations Manager**

Method Validation Report

Test Method: Determination of Volatile Matter

Standard Method: ISO 562

Matrix: Coal samples

Laboratory: AzureSky

Analyst: Keitumetse Maleka

Date: 06 May to 23 June 2025

1. Purpose

To validate the laboratory's method for determining volatile matter in coal in accordance with ISO 562, ensuring the method is fit for its intended use and meets ISO/IEC 17025 requirements.

2. Scope

This validation covers volatile matter determination for various coal types received in the laboratory, in compliance with ISO 562.

3. References

- ISO 562: Hard coal and coke — Determination of volatile matter

- ISO/IEC 17025:2017

- Laboratory Quality Manual

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4. Validation Parameters & Results

Parameter	Requirement	How to Evaluate	Acceptance Criteria	Results
Repeatability	Precision under same conditions	10 replicates on same sample	%RSD<0.3. 3% of the mean, for vol values >10%	• Repeatability:0.417 % • %RSD values 0.518 and 0.484, for analysts 1 and 2; respectively.
Reproducibility	Precision across analysts/instruments/ days	Same sample by 10 different PT participants	0.5% or 4% of the mean, whichever is greater.	0.327%
Accuracy	Comparison to CRM or known value	Test CRM with known volatile matter	Recovery 98–102%	99.22%
LOD	Not typically needed for VM	Not applicable for gravimetric % methods	–	–
Robustness	Small variations in time/temp/sample weight	Change time from 7 min to 4 min	Difference ≤ 0.3% VM	0.39
Uncertainty	Required by ISO/IEC 17025	Use GUM approach or Nord	k=2, 95% confidence	0.069

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		test Guide		

5. Equipment Used

Equipment	Model/ID	Calibration Due
Furnace	AS01338	March 2026
Analytical Balance	AS2156	March 2026
Crucibles with lids	ISO compliant	–

6. Sample Types Validated

- Bituminous coal

7. Validation Summary

The volatile matter determination method has demonstrated acceptable precision, accuracy, and ruggedness in accordance with ISO 562. However, the robustness has failed. **This could be because the running times were too different (4 and 7 min).** Measurement uncertainty has been calculated and documented.

8. Conclusion

The method validation fails due to failure in the statistical evaluation of robustness. Thus, the method is not suitable for its purpose. Analysis shall be repeated for 6 and 8 min; these will be compared with the 7 min standard time. Re-evaluation of the results is documented below.

06/2025	00	06/2028

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9. Robustness re-evaluation

t-Test: Paired Two Sample for Means

	7 MIN	8 MIN
Mean	28.506	28.612
Variance	0.0223175	0.0095
Observations	5	5
Pearson Correlation	0.27339280	
Hypothesized Mean Difference	4	
df	0	
t Stat	4	
P(T<=t) one-tail	-1.53421915	
t Critical one-tail	9	
P(T<=t) two-tail	0.09988048	
t Critical two-tail	6	
	0.19976097	
	2	
	2.77644510	
	5	

Evaluation	
Ho:	There is no significant difference b/w the means of 7- and 8-min results
Ha:	There is a significant difference b/w the means of 7- and 8-min results
tcalculated:	0.09988048
p-value (2 tail) :	0.19976097

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	2
	2.77644510
tcritical	5

Since the p-value is greater than 0.05, and the tcalculated is less than tcritical.
Accept the null hypothesis.

Conclusion: There is no significant difference b/w the 7 min and 8 min. This implies that for volatile analysis, samples left in the furnace for 7-8 min would produce results that are close to each other.

LIMS NO	6 MIN	7 MIN	8 MIN	LIMS NO
1268	28.57	28.65	28.76	1273
1269	28.56	28.43	28.56	1274
1270	28.55	28.345	28.66	1275
1271	28.26	28.425	28.52	1276
1272	28.31	28.68	28.56	1277

t-Test: Paired Two Sample for Means

	7 MIN	*6 MIN*
Mean	28.506	28.45
Variance	0.022175	0.02305
Observations	5	5
Pearson Correlation	0.178014057	
Hypothesized Mean Difference	0	
df	4	
t Stat	0.541663593	
P(T<=t) one-tail	0.308405515	
t Critical one-tail	2.131846786	
P(T<=t) two-tail	0.61681103	

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t Critical two-tail	2.776445105

Evaluation

Ho:	There is no significant difference b/w the means of 6- and 7-min results
Ha:	There is a significant difference b/w the means of 6- and 7-min results
tcalculated:	0.541663593
p-value (2 tail) :	0.61681103
tcritical	2.776445105

Since the p-value is greater than 0.05, and the tcalculated is less than tcritical. Accept the null hypothesis.

Conclusion: There is no significant difference b/w the 7 min and 6 min. This implies that for volatile analysis, samples left in the furnace for 7-6min would produce results that are close to each other.

10. Glossary

Terms to know and why:

1. **Robustness**: The method's ability to remain unaffected by small, deliberate changes in method parameters- such as: Holding time, furnace temperature, Sample mass, lid coverage or crucible type.

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Why is it important: In real laboratories slight deviations do happen, instruments may drift slightly, or timing may vary b/w runs. You want to know if the method can tolerate those changes without producing significant errors. This means that results are dependent on operator precision or timing, which introduces systematic errors.